

ARISTOTLE UNIVERSITY OF THESSALONIKI
FACULTY OF ENGINEERING – SCHOOL OF MECHANICAL ENGINEERING
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SEMESTER: 8th
COURSE: DATA ANALYSIS

3rd ASSIGNMENT

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Introduction

The third assignment uses machine learning and its various models in order to draw conclusions regarding the question of which of the predictor variables (other than quality) can predict or estimate, to a greater degree, the wine type (white, red).

1. Part A

In the first part of this assignment, the estimation/prediction of the wine type from the predictor variables Alcohol (alcohol), pH and Chlorides (Chlorides) is requested, using the following methods for each one separately:

- Logistic Regression (LR)
- Linear Discriminant Analysis (LDA)
- Naive Bayes (NB), with the assumption that the class characteristics follow a normal distribution.

Initially, the first lines of the code import the libraries that will be used (numpy, pandas, matplotlib and seaborn) and, from the integrated “sklearn”, the models that will be used. Then the file with the “clean” data is loaded, from which we obtain information about the quantity of white and red wines. The variable that will be set as the target is “wine_type”, while at the same time, together with “quality”, it will be excluded from the predictors.

Next, we define the classes as red: 0, white: 1. From sklearn, the function “train_test_split” was imported, through which the data will be split into training and test data. The two variables contain 70% and 30% respectively. In addition, the parameters “random_state” (based on the AEM = 7137) and “stratify” are introduced, with the help of which the equal split of the white/red types across the two variables is achieved, with respect to their original distribution.

The initial dataset had shape (rows, columns): (5295, 13), while the training set: (3706, 13) and the test set: (1589, 13). The images below show how many of the records belong to each of the two categories.

```
wine_type distribution:
wine_type
white    3991
red      1304
Name: count, dtype: int64
```

Image 1

```
Train wine_type distribution:
wine_type
white    2793
red       913
Name: count, dtype: int64
```

Image 2

```
Test wine_type distribution:
wine_type
white    1198
red       391
Name: count, dtype: int64
```

Image 3

In the code, the variables x_train, x_test, y_train, y_test appear. The x variables contain the values of the predictors, while the y variables contain the actual response values. “Train” and “test” indicate whether they are used in training the model or in testing it. The evaluation of the model then begins. This evaluation will be based on various measurements, to finally select both the best prediction variable and the best model.

First, the training of the model on the training data begins, learning the relationship between the predictor X and the response Y. Then, the prediction on the x_test data follows, and the model's responses are stored in the variable y_pred, so that they can finally be compared with the actual wine types stored in y_test.

The comparison is carried out with 5 different indicators. The Confusion Matrix presents how many values the model placed in each type and how many belonged there. In this indicator, the variables TN (true negative), FP (false positive), FN (false negative) and TP (true positive) also appear, with positive meaning the value 1 -> white and negative the value 0 -> red. In this way it records the total of the model's successes and failures.

Then follows the Accuracy of the model, the quotient (correct responses / total). Next, the Precision indicator, which measures how many of those it guessed as white were actually white (pos_label = 1); then the Recall indicator, which measures how many of the actual whites it found; and finally, the F-1 score indicator, which combines the precision and recall indicators. Below, the indicators for each predictor and model are presented separately.

VARIABLE: alcohol	VARIABLE: pH	VARIABLE: chlorides
<pre> --- Logistic Regression --- Confusion Matrix: [[0 391] [0 1198]] Accuracy : 0.7539 Precision: 0.7539 Recall : 1.0000 F1-score : 0.8597 --- LDA --- Confusion Matrix: [[0 391] [0 1198]] Accuracy : 0.7539 Precision: 0.7539 Recall : 1.0000 F1-score : 0.8597 --- Naive Bayes --- Confusion Matrix: [[0 391] [0 1198]] Accuracy : 0.7539 Precision: 0.7539 Recall : 1.0000 F1-score : 0.8597 </pre>	<pre> --- Logistic Regression --- Confusion Matrix: [[39 352] [40 1158]] Accuracy : 0.7533 Precision: 0.7669 Recall : 0.9666 F1-score : 0.8552 --- LDA --- Confusion Matrix: [[41 350] [47 1151]] Accuracy : 0.7502 Precision: 0.7668 Recall : 0.9608 F1-score : 0.8529 --- Naive Bayes --- Confusion Matrix: [[47 344] [51 1147]] Accuracy : 0.7514 Precision: 0.7693 Recall : 0.9574 F1-score : 0.8531 </pre>	<pre> --- Logistic Regression --- Confusion Matrix: [[18 373] [12 1186]] Accuracy : 0.7577 Precision: 0.7607 Recall : 0.9900 F1-score : 0.8604 --- LDA --- Confusion Matrix: [[60 331] [24 1174]] Accuracy : 0.7766 Precision: 0.7801 Recall : 0.9800 F1-score : 0.8687 --- Naive Bayes --- Confusion Matrix: [[73 318] [28 1170]] Accuracy : 0.7823 Precision: 0.7863 Recall : 0.9766 F1-score : 0.8712 </pre>

Image 4

Image 5

Image 6

First of all, it is observed that all three models that ran with the prediction variable alcohol fail to register even a single record as TN (i.e. red); at the same time, these values have all

been registered as white, since all the values of the red wines are declared as FP. In this way, all the indicators for this particular predictor remain constant across the three models. This is reasonable, since the values estimated for both the white and the red wines do not differ.

Next, it is found that the prediction variable pH can better distinguish and separate the reds from the whites, since records have been placed both in TN and in TP. It is observed, of course, that there are records of whites that were incorrectly registered as reds. The sum of the incorrect placements (LR: 392, LDA: 397, NB: 395) appears to exceed the value of 391 of the alcohol models. That is, in the attempt to separate the whites from the reds, more errors were made than if they had all been registered as whites. This is also perceived from the Accuracy indicator, which for all the pH models appears to be lower than for all the alcohol models. Although pH shows slightly lower accuracy, its behavior is more realistic, as it attempts a real separation of the categories.

With a first glance at the confusion matrix for the prediction variable Chlorides, it appears that two of the three models have been trained quite well, so as to understand the relationship between x and y. This is perceived from the increased records (LDA: 60, NB: 73) as reds (TN) compared both to the previous variables and to the Logistic Regression model, which ranges at the levels of the pH models. This increase is not accompanied by a simultaneous increase of the FN (24, 28 respectively), a fact that proves the importance of the chlorides variable, as well as the models' understanding of the system. Finally, Naive Bayes appears to outperform the LDA method in terms of accuracy (Accuracy: 0.782 vs 0.777), precision (Precision: 0.786 vs 0.780) and F1-score (F1-score: 0.871 vs 0.867).

The Recall indicator of the LDA method appears to be higher, but this happens because it found more whites (1174 vs 1170) and only that, since the indicator measures the percentage of whites found out of their total. For this reason, in the variable alcohol the Recall indicator of the three methods equals one (1.00), since all the whites were placed correctly, regardless of whether these records also contained reds as incorrect placements.

PART A SUMMARY TABLE									
Feature	Method	Accuracy	Precision	Recall	F1-score	TN	FP	FN	TP
alcohol	Logistic Regression	0.753933	0.753933	1.000000	0.859706	0	391	0	1198
alcohol	LDA	0.753933	0.753933	1.000000	0.859706	0	391	0	1198
alcohol	Naive Bayes	0.753933	0.753933	1.000000	0.859706	0	391	0	1198
pH	Logistic Regression	0.753304	0.766887	0.966611	0.855244	39	352	40	1158
pH	LDA	0.750157	0.766822	0.960768	0.852908	41	350	47	1151
pH	Naive Bayes	0.751416	0.769282	0.957429	0.853105	47	344	51	1147
chlorides	Logistic Regression	0.757709	0.760744	0.989983	0.860355	18	373	12	1186
chlorides	LDA	0.776589	0.780066	0.979967	0.868664	60	331	24	1174
chlorides	Naive Bayes	0.782253	0.786290	0.976628	0.871184	73	318	28	1170

Image 7

In conclusion, the variable that “fits” best in the system is Chlorides, as it manages to have the highest values of the Accuracy indicator regardless of method, proving that the predictions from this variable are more correct, more accurate and lead to a greater number of successful registrations to the correct wine type.

However, based on the results of the chlorides variable, no substantial difference is observed in performance among the three classification methods. All three methods present similar levels of precision and very high recall values, which indicates that they are equally capable of recognizing the positive class (white wines). Naive Bayes shows marginally the best overall performance (Accuracy ≈ 0.782 and F1-score ≈ 0.871), followed by LDA, while Logistic Regression shows slightly lower performance. We therefore reach the conclusion that, except for Logistic Regression — which appears to have a slightly weaker discriminative ability between the wine types — there is no significant difference in their indicators.

```
=====
SORT BY ACCURACY
=====
```

Feature	Method	Accuracy	Precision	Recall	F1-score	TN	FP	FN	TP
chlorides	Naive Bayes	0.782253	0.786290	0.976628	0.871184	73	318	28	1170
chlorides	LDA	0.776589	0.780066	0.979967	0.868664	60	331	24	1174
chlorides	Logistic Regression	0.757709	0.760744	0.989983	0.860355	18	373	12	1186
alcohol	Logistic Regression	0.753933	0.753933	1.000000	0.859706	0	391	0	1198
alcohol	LDA	0.753933	0.753933	1.000000	0.859706	0	391	0	1198
alcohol	Naive Bayes	0.753933	0.753933	1.000000	0.859706	0	391	0	1198
pH	Logistic Regression	0.753304	0.766887	0.966611	0.855244	39	352	40	1158
pH	Naive Bayes	0.751416	0.769282	0.957429	0.853105	47	344	51	1147
pH	LDA	0.750157	0.766822	0.960768	0.852908	41	350	47	1151

Image 8

2. Part B

In the second part of the assignment, it is requested to apply for the whole set of predictor variables (predictors), the following methods for estimating the wine type:

- K-nearest-neighbors (KNN),
- Logistic Regression,
- Linear Discriminant Analysis (LDA),
- Quadratic Discriminant Analysis (QDA),
- Naïve Bayes (NB), with the assumption that the class characteristics follow a normal distribution.

The predictors this time are all the variables mentioned in the given file, except wine type and quality. That is, they are: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol. The models will be trained on the whole set of training data (which contains the values for all the variables) and will try to “understand” the relationships between the values of the variables and the binary response variable wine type. In continuation of Part A of the assignment, the red type is symbolized as negative (negative, 0) and the white type as positive (1).

The training and test sets remain as they are, with Train shape: (3706, 13) and Test shape: (1589, 13), and with white/red values shown in Image 2 and Image 3. However, some models may need normalization of the data so that their processing is possible; for this reason, from the library “sklearn.preprocessing” we import “StandardScaler”. With the help of this library, a new set is created for the training data (x_train_scaled) and the test data (x_test_scaled). The

normalization (scaling) of the data was carried out for the methods that are based on distances or optimization procedures, such as KNN and Logistic Regression. This process transforms the variables so that they have a mean of 0 and a standard deviation of 1, ensuring that all the variables contribute equally to the model. Without the scaling, the variables with a larger range of values would dominate the calculation of distances or coefficients, thus leading to misleading results.

For the further evaluation of the models and the construction of the ROC curves, the calculation of a continuous prediction measure (score) is required, and not just the final binary response (0 or 1). For this reason, the code checks whether each model has the `predict_proba` method, in which case the probability of the positive class (white) is used. Otherwise, it checks whether the `decision_function` method exists, which returns a relative decision score. If neither of the two is available, the calculation of the ROC (Receiver Operating Characteristic) curve and the AUC (Area Under Curve) value is not possible. This process is necessary, since the ROC curve and the AUC index are based on continuous prediction values and allow the overall estimation of the model's discriminative ability regardless of the classification threshold.

Before the methods “run”, the number K of checked neighboring values must be appropriately defined for KNN. In the problem statement, it is requested to select it with the 5-Fold Cross-Validation method. This method splits the data into 5 pieces, trains with the K number on 4 of them and tests on the 5th. This process is repeated so that all 5 segments pass through the test position. The K number with the highest accuracy is selected for the KNN model. In this case, the most appropriate K (“Best K”) is 7, with Accuracy: 0.9682. Thus, the KNN model will decide based on the 7 nearest neighbors.

```
=====
SELECTION OF BEST K FOR KNN
=====
Best K: 7
Best CV Accuracy: 0.9682
```

Image 9

Finally, the models and the data sets they will use are selected. The KNN and LR models, as mentioned previously, will “run” with normalized data, while the remaining models with the normal set. As the models run, they also produce and print the indicators required for their evaluation. These indicators are those used in Part A (Confusion Matrix, Accuracy, Precision, Recall, F-1 score) with the addition of the AUC. The AUC (0.0–1.0) measures the area under the ROC curve and interprets the ability of the method to distinguish the 2 classes overall.

Finally, running the program and after the indicators are calculated, the summary table is created, where the following are depicted: the method, Accuracy, Precision, Recall, F-1 score, AUC, TN, FP, FN, TP. At first glance, it appears that the KNN method outperforms the rest, as it presents higher Accuracy (0.973), Precision (0.982) and AUC (0.973), more TN (369) — equal to LDA — but also fewer FN (21). Nevertheless, with the exception of the QDA and Naïve Bayes methods, which present quite lower Accuracy values (0.925 and 0.948 respectively), the methods (KNN, LR, LDA) do not appear to have significant differences in

their performance. In closing, the ROC curves are presented, both overall in Image 10 and separately, each one, in the Appendix.

PART B SUMMARY TABLE									
Method	Accuracy	Precision	Recall	F1-score	AUC	TN	FP	FN	TP
KNN	0.972939	0.981651	0.982471	0.982061	0.972690	369	22	21	1177
Logistic Regression	0.971051	0.980000	0.981636	0.980817	0.966173	367	24	22	1176
LDA	0.972310	0.981636	0.981636	0.981636	0.965877	369	22	22	1176
QDA	0.925110	0.954507	0.945743	0.950105	0.941727	337	54	65	1133
Naive Bayes	0.947766	0.968094	0.962437	0.965257	0.954509	353	38	45	1153

Image 10

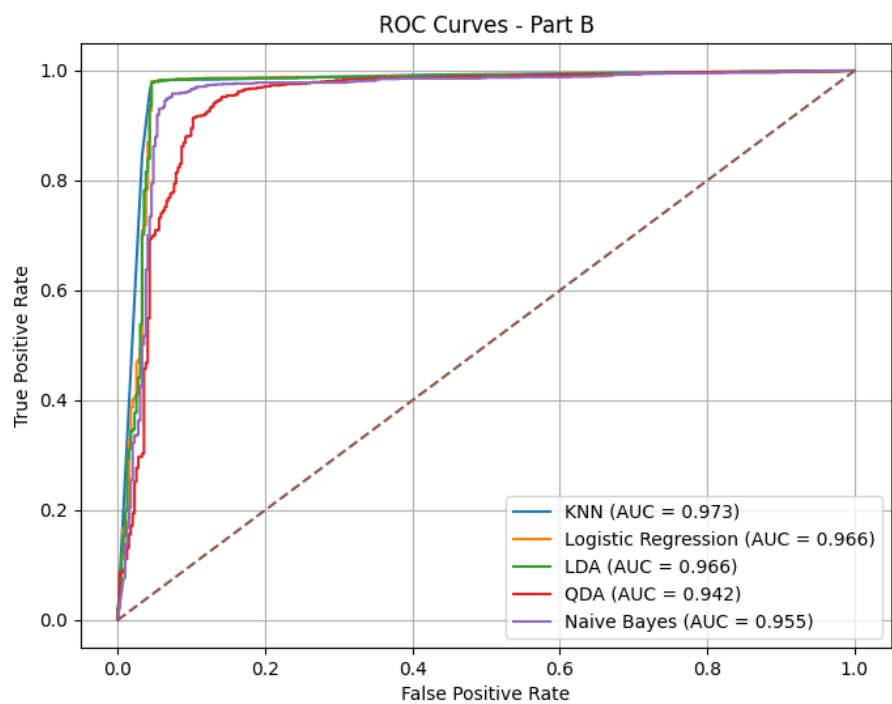


Image 11

Appendix

The individual ROC curves for each method are collected here. (The original numbering in this section was inconsistent; the placeholders below are labeled by model for clarity.)

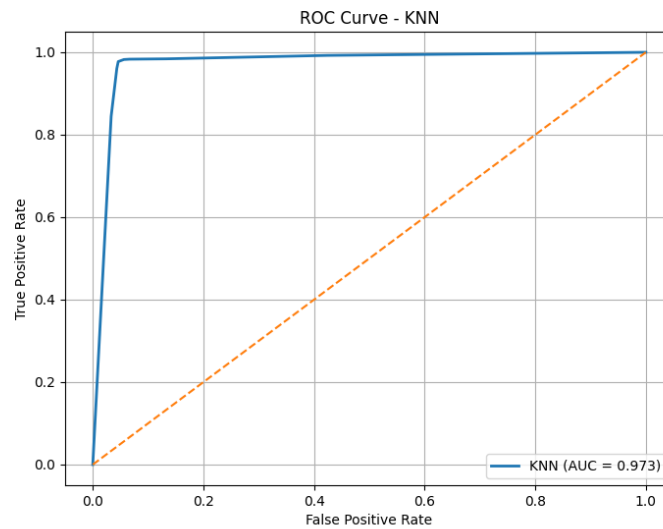


Image 11

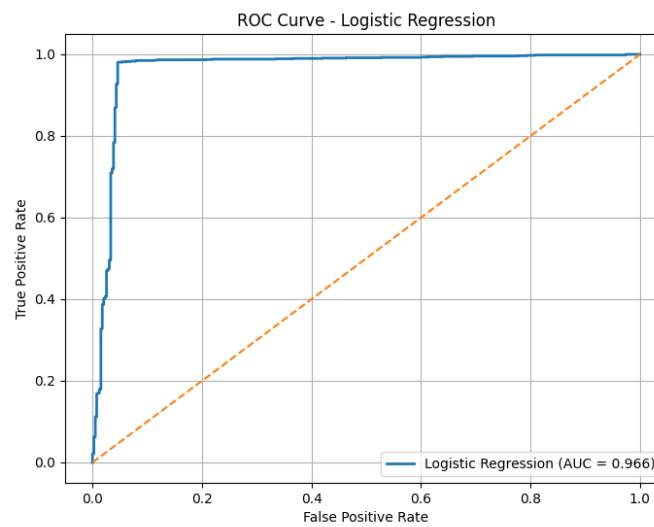


Image 12

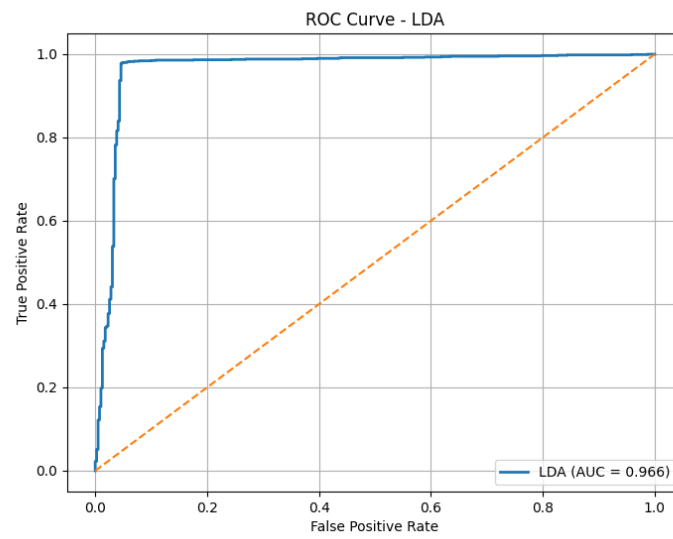


Image 13

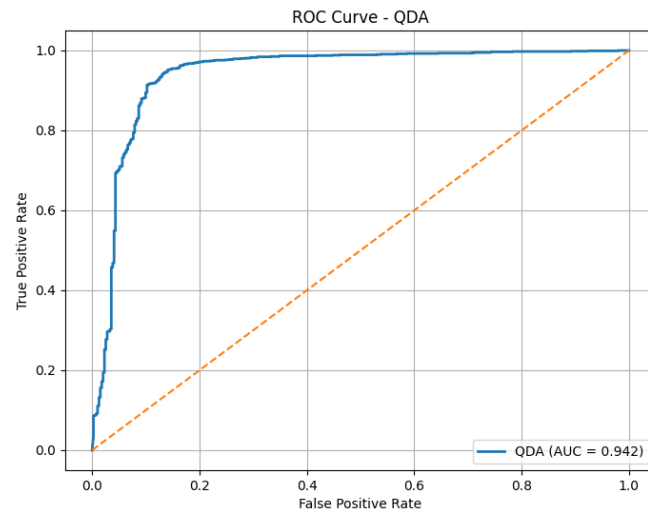


Image 14

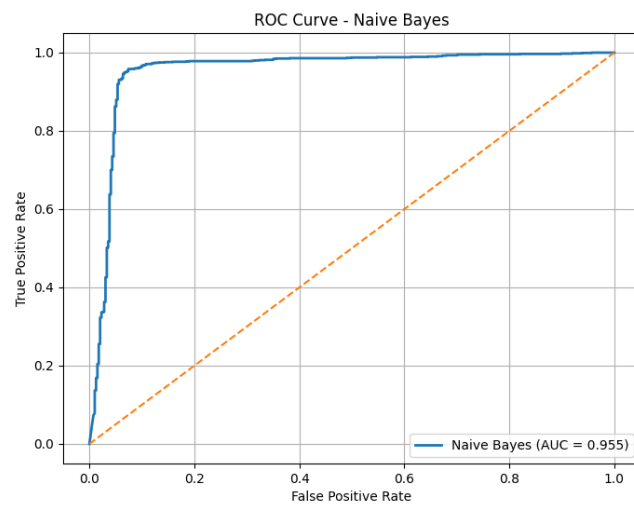


Image 15